

Sara Ichinaga

I am an applied mathematician with extensive experience with data analysis, mathematical modeling, and machine learning. I am interested in developing data-driven algorithms and software for applications across various scientific and engineering disciplines.

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SUMMARY OF QUALIFICATIONS

- Ph.D. candidate in applied mathematics with a focus on algorithm development for data-driven modeling.
- Python expert with experience developing software packages and collaborating on open-source code.
- Experience presenting high-level mathematics and machine learning concepts to various audiences.

EDUCATION

Ph.D. Applied Mathematics, GPA 3.92/4.0, University of Washington, Seattle. **2021-Present**

Advisors: Prof. J. Nathan Kutz and Prof. Steven L. Brunton. Expected June 2027.

M.S. Applied Mathematics, GPA 3.87/4.0, University of Washington, Seattle. **2021-2022**

B.S. Applied and Computational Mathematics, GPA 3.94/4.0, University of Washington, Seattle. **2017-2021**

RELEVANT COURSEWORK

- (CSE 546) Machine Learning
- (CSE 547) Machine Learning for Big Data
- (AMATH 515) Optimization: Fundamentals and Applications
- (AMATH 582) Computational Methods for Data Analysis
- (AMATH 584) Numerical Linear Algebra

SELECTED PROFESSIONAL EXPERIENCE

Graduate Research and Teaching Assistant **Sept 2021-Present**

Department of Applied Mathematics, University of Washington, Seattle

- Developed two extensions of the dynamic mode decomposition (DMD): (1) sparse-mode DMD for learning spatially local and global structures from data, and (2) mrCOSTS for decomposing multi-scale data.
- Formulating new noise-robust weak form methods for DMD with control (DMDc).
- Developer and maintainer for the open-source Python package PyDMD: <https://github.com/PyDMD/PyDMD>.
- Teaching assistant for ENGR 510, an introductory machine learning course for engineering professionals.

Ph.D. Applied Scientist Intern, Roblox, Distributed Physics **Summer 2026**

- Researched the extent at which reduced order models (ROMs) could be used for modeling and simulating positions, orientations, and velocities of various in-game mechanisms.
- Incorporated online learning methods into the game engine for real-time ROM computation and utilization.

Computational Software Development Intern, National Laboratory of the Rockies (NLR) **Summer 2025**

- Developed PyRidge, an NLR Python package for data-driven surrogate modeling, active subspace learning, and dimension reduction via ridge function approximation: <https://github.com/NatLabRockies/PyRidge>.
- Applied PyRidge to NLR data to determine under-sampled regions of design parameter space.

SELECTED PUBLICATIONS AND PREPRINTS

- [Sara M. Ichinaga](#), Steven L. Brunton, Aleksandr Y. Aravkin, and J. Nathan Kutz. "Sparse-mode dynamic mode decomposition for disambiguating local and global structures." *In review, SIAM Journal on Applied Dynamical Systems*. 2025. [\[preprint\]](#)
- [Sara M. Ichinaga](#), Francesco Andreuzzi, Nicola Demo, Marco Tezzele, Karl Lapo, Gianluigi Rozza, Steven L. Brunton, and J. Nathan Kutz. "PyDMD: A Python package for robust dynamic mode decomposition." *JMLR*. 2024. **25**(417):1-9. [\[paper\]](#) [\[long version on arXiv\]](#)
- Karl Lapo, [Sara M. Ichinaga](#), and J. Nathan Kutz. "A method for unsupervised learning of coherent spatiotemporal patterns in multi-scale data." *PNAS*. 2024. **122**(7):e2415786122. [\[paper\]](#)

TECHNICAL SKILLS

Coding Languages: Python (expert), MATLAB, Bash, Java, SQL, R

Python Libraries: NumPy, Matplotlib, SciPy, PyTorch, Scikit-Learn, Pandas, CVXPY, PyTest

Technologies: Git/GitHub, Jupyter, Anaconda, LaTeX, Microsoft Azure